

Quality Management System Manual

Example LLC

ISO 9001:2015 · Stage 1 Foundation

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1. Quality Policy

At Example LLC, every piece of measuring or test equipment that leaves our laboratory has been calibrated strictly against reference standards that are directly traceable to NIST. Our calibration certificates precisely reflect the actual measurement results obtained for each item — we do not estimate or round values, and no certificate is issued without a complete and auditable trace of what was performed. If there is any doubt about a result, we do not associate our name or documentation with that work.

We recognize that the integrity of our process is frequently scrutinized not only by our direct customers, but often by third parties via audits or their customers' own quality processes, especially in automotive and aerospace supply chains where traceability is critical. Therefore, our documentation systems and procedures are designed so that any certificate can be pulled at random and the full calibration process, including equipment traceability, will withstand external examination.

Each calibration is executed using NIST-traceable reference standards. These standards are themselves managed with clear scheduling to ensure timely recalibration via qualified external providers, and are always logged for current status. Our workflow tracks the use of each reference standard against specific jobs to ensure traceability and to identify potential impacts if a standard is discovered out of tolerance.

We are committed to turnaround times that support our customers' production requirements. Our job scheduling prioritizes work order clarity at intake, confirms delivery expectations, and regularly monitors job progress to minimize delays. Whenever specification errors or delays occur, we review each incident for root cause and corrective action. All such complaints and corrections are logged, analyzed, and used to reinforce staff training and documentation.

We operate under a strict policy of not cutting corners on documentation or traceability, driven by our belief that the only value of calibration is confidence in the result. Any process or staff deviation is treated as a learning opportunity for improvement, and our procedures, training, and systems are regularly adapted for clarity and consistency across the laboratory team.

2. Quality Objectives

Our current quality objectives are defined with measurable targets and are reviewed and adjusted annually as part of the management review process:

- **Certificate Correction Rate:** As of 2025, 6 calibration certificates required post-issue corrections, most commonly due to unit errors or missing measurement values. Our objective is to reduce this to 2 or fewer corrections per year by the end of 2026. Progress is tracked monthly. Root cause for each correction is investigated, and corrective actions are implemented and documented.
- **On-Time Delivery Rate:** The current on-time delivery rate is approximately 88%. Our goal is to achieve 95% or above on-time delivery by the end of 2026. Delays are tracked in the lab management software and analyzed during monthly planning meetings. Backlogs or any single job exceeding promised turnaround triggers review for workflow or scheduling issues.
- **Reference Standards Recalibration:** In 2025, 2 reference standards became overdue, which is unacceptable for traceable calibration work. Our objective from Q1 2026 is to maintain 0 overdue reference standards at any time. Due dates for all standards are tracked continuously, with 60-day advance alerts for scheduling external recalibrations. An overdue status triggers immediate investigation and potential customer notification if any affected calibrations are identified.

These objectives are directly linked to operational priorities: error prevention, customer confidence in documentation, and maintaining uninterrupted, standards-based calibration service.

3. Scope of the Quality Management System

The quality management system of Example LLC covers all calibration and test measurement services performed at our main laboratory in Greenville, South Carolina, as well as on-site calibration provided at customer premises. This includes the calibration of dimensional, pressure, temperature, and electrical equipment such as gauges, torque tools, multimeters, and related instruments — for industrial manufacturers in regulated supply chains (primarily automotive and aerospace).

The system encompasses the entire process life cycle: customer inquiry, job intake and scheduling, calibration and measurement activities, certificate generation, shipping/return, reference standard management, complaint and corrective action processing, document control, and supplier management for critical calibration services and shipping. All laboratory personnel, administrative staff involved in calibration-related activities, and external providers responsible for NIST-traceable calibrations of our reference standards fall within the system boundaries.

Excluded: The QMS does not extend to manufacturing or testing performed by customers, or to activities outside calibration or measurement traceability.

4. Process Overview

All key laboratory workflows are organized into leadership, core, and support processes that together ensure error-free calibration, documentation, and timely delivery of service.

4.1 Leadership Processes

Leadership processes are directed by Dana Whitfield (Owner) and are designed to set and maintain the laboratory's quality standards and oversee the operational structure.

Dana reviews and approves all changes to quality policy, procedural documentation, and leads the annual management review. Key information for review is gathered from lab error/correction logs, delivery statistics, and reference standard tracking. Dana also provides final approval for corrective actions that involve major process or personnel changes.

The leadership cycle includes initial definition and periodic review of the company's quality policy and objectives, the setting of annual operational targets, and monitoring compliance with all relevant regulatory requirements, including NIST traceability and OSHA safety. When significant deviations from key objectives occur — such as repeated corrections or missed delivery targets — Dana reviews investigation reports, approves required resources for remedial training or system improvements, and sets expectations for future internal audits.

In cases where reference standards are delivered back out-of-tolerance, Dana is involved in the risk assessment (with Marcus Bell) to determine whether prior customer jobs require re-evaluation or notification. Supplier reliability or issues with external calibration partners are reviewed by Dana as part of a supplier management process.

4.2 Core Processes

Lab Calibration Workflow:

The calibration process begins when a customer contacts the company via phone, email, or online form. Theresa Nguyen receives the inquiry, logs the job into the system, confirms with the customer the precise scope of calibration work, required documentation (including, if applicable, customer-specific format or data requirements), and expected turnaround.

When equipment arrives, Theresa checks the delivery against the job entry, inspects items for damage or discrepancies, confirms identification, and notifies Marcus Bell. If equipment is scheduled for on-site calibration, Theresa coordinates logistics for the site visit and ensures all necessary reference standards and supporting tools are packed for travel.

Marcus Bell reviews the job scope, assigns work to the technician with appropriate specialization (dimension, temperature, pressure, or electrical). Before starting, the technician verifies the reference standards assigned for each measurement, checks their current calibration status (due dates and records), and confirms applicability for the specific equipment type.

The technician performs all calibrations following documented procedures, entering raw measurement data both in bench notes (physical or digital as appropriate) and directly into the lab management software. Where customer-specific requirements exist, the technician uses predefined certificate templates. On completion, the technician labels the equipment with a dated calibration sticker referencing the certificate number.

Certificates are subjected to review by Dana or Marcus. This can be review of every certificate or periodic spot checks depending on workload and personnel availability. Any discrepancies, missing values, or uncertainties are returned to the responsible technician for correction before the certificate is finalized.

After approval, Theresa packages the equipment and prepares outbound shipment with the corresponding certificate. If calibration is performed on-site, the responsible technician delivers the equipment and documentation directly to the customer contact and ensures receipt acknowledgment.

Reference Standard Management:

Kevin Park monitors all in-house reference standards via a digital tracking log. Sixty days before any standard's recalibration is due, the system flags the item and Kevin initiates its send-out to a pre-approved, accredited external provider. Documentation for the standard (such as previous certificates and current usage logs) is included with the shipment.

Upon return, Kevin receives the recalibrated standard and certificate, compares returned values with expected tolerance, and enters the new data in both the tracking log and any linked equipment records. If any standard comes back out-of-tolerance, Kevin immediately escalates to Marcus and Dana for assessment. Together, they review which jobs, if any, were performed using the affected standard during the last calibration cycle. If there is any risk to prior measurement validity, customers are notified and impacted jobs recalculated or re-certified as necessary.

Complaint and Certificate Correction:

When a customer reports a problem with a certificate — incorrect value, missing unit, or omission — Theresa logs the complaint in the system and informs Dana for rapid resolution. Dana pulls original measurement records and assigns the review to the responsible technician, who cross-checks bench records and the certificate. If a procedural or data entry error is discovered, a corrected certificate is generated. The correction includes a narrative explaining the change and is delivered to the customer.

Where the error was due to system or training gaps, Marcus investigates, identifying if a process procedure needs updating or whether further employee training is warranted. If patterns or repeated problems emerge from corrective action logs, a formal root cause investigation is initiated by Marcus, and Dana is notified for oversight and approval of action plans. The process remains open until documented closure and follow-up evaluation.

4.3 Support Processes

Document and Record Management:

All calibration-related documentation, including customer jobs, bench notes, certificates, reference standard logs, and corrective action records, are managed through our central lab management software. Theresa and Marcus ensure that all entries are made in real time and that only current, approved versions of certificate templates or procedures are used.

Regular system backups are scheduled, and access is restricted according to user role. Certificate templates and procedural documents are periodically reviewed and updated under Marcus's supervision. Any change requires review and approval by Dana, with all revisions tracked via controlled version numbers and review logs.

Supplier Management:

Suppliers for external calibration of reference standards are vetted for accreditation status and reliability. Kevin monitors their calibration certificates and turnaround history. If an issue arises with supplier performance or documentation, Dana initiates a review for alternative sourcing or additional supplier oversight.

Staffing and Training:

All technicians receive baseline training on documented procedures for each calibration type and are assigned jobs matching their technical background and experience. Whenever a corrective action identifies a training need, Marcus develops targeted sessions, and attendance is recorded. New technicians or trainees are supervised, with records of completed checklists and reviews maintained.

Administrative Support:

Theresa is responsible for front office operations, customer invoicing, logistics coordination for incoming and outgoing shipments, and support for job scheduling. She ensures detailed information for customer and shipping records, tracks equipment during transit, and updates customers in case of any delay.

5. Roles and Responsibilities

Dana Whitfield, as Owner, holds final responsibility for overall quality, resource allocation, and ensuring that the QMS meets internal goals and all customer/stakeholder requirements. Dana approves all policies, chairs management reviews, and directly intervenes in escalated quality or supplier incidents. Dana is the principal authority for determining customer notification policies when nonconforming calibrations are discovered.

Marcus Bell, Lab Manager and Quality Manager, is responsible for day-to-day operational execution of the calibration process, supervision of all technicians, routine review of calibration certificates, and the maintenance of discipline-specific procedures. Marcus initiates root cause analyses for any repeated problems, tracks the closure of corrective actions, and oversees practical training for both new and experienced technicians. Marcus is the primary contact for technical escalations and is empowered to update procedures as necessary, subject to final approval by Dana.

Theresa Nguyen serves as the front office coordinator, managing job intake, scheduling, inbound and outbound logistics, and customer billing. Theresa also records customer feedback and complaints, provides status updates to customers, and maintains up-to-date shipping and job records in the lab management software.

Kevin Park is responsible for the ongoing tracking of all reference standards, scheduling recalibrations, and ensuring timely communication with and evaluation of external suppliers. Kevin prepares standards for shipment, logs returned certificates, and immediately notifies Marcus and Dana if any discrepancies or out-of-tolerance results are encountered.

Bench Technicians (nine staff and one trainee) execute calibrations according to assigned work orders and approved procedures for their area of specialization. They complete bench notes and ensure calibration results are thoroughly recorded and traceable to qualified reference standards. The trainee operates under close supervision with documented signoffs to confirm procedural adherence.

Interaction among staff is routine and direct, with Theresa and Marcus acting as central communication points. Escalations or process changes involving quality or customer impacts require Marcus's involvement, with Dana making final decisions if broader system or external communication is involved.

6. Document Control Procedure

All documented information related to the QMS, including policies, procedures, calibration protocols, certificate templates, and records, are maintained within the lab management software or secured shared drives. Each controlled document is identified with a revision number, effective date, and approved by Dana Whitfield.

Requests to update or revise procedures can be initiated by any staff member and submitted to Marcus Bell for evaluation. Drafts are circulated electronically for comment, and any substantial change, especially changes involving measurement methodology or documentation format, requires written approval from Dana. Once finalized, prior versions of the document are archived in a restricted access directory and clearly marked as obsolete.

Periodic document reviews occur at least annually. Any emergency changes resulting from corrective actions or external audit findings are updated immediately, followed by training or briefing for affected staff. Only current versions of documents are accessible to personnel for daily operations; access to edit or approve documentation is limited to Marcus and Dana.

Records of calibration, bench notes, certificates, reference standard logs, and corrective action documents are retained for a minimum of ten years, or longer if specified by contract or regulation. Disposal of records after retention period requires approval by Marcus.

7. Internal Audit Procedure

Internal audits are scheduled at least annually, with Marcus Bell responsible for audit planning and execution. Audits cover all core and support processes, including sample reviews of calibration records, certificate accuracy, reference standard trace log integrity, and compliance with document control requirements.

Each audit involves targeted checks of randomly selected jobs and documents, interviews with technicians, and evaluation of practical procedure adherence. The auditor records all findings, identifies any nonconformities or areas requiring improvement, and discusses preliminary results with Dana.

Findings demanding immediate attention (such as systemic documentation errors or overdue reference standards) trigger corrective actions that are tracked until closed. After each audit round, Marcus summarizes results and presents them at the management review for confirmation, further planning, or resource allocation as needed.

Audit records, including checklists, reports, findings, and closure evidence, are retained and subject to customer or regulatory review upon request.

8. Management Review Procedure

A full management review is conducted annually, led by Dana Whitfield and with participation from Marcus Bell and relevant supporting staff. The agenda is set in advance and includes:

- Review of the quality policy and current objectives, comparing target achievement with ongoing performance metrics.
- Analysis of audit findings, customer complaint logs, system corrections, and corrective actions resolved and outstanding.
- Evaluation of supplier performance, particularly external calibration providers, with any recommended changes for the approved provider list.
- Review of training completeness, technician qualification updates, and adequacy of procedure documentation.
- Assessment of resource needs for potential workload increases or technology upgrades.

During the review, gaps or shortfalls are identified with proposed plans for improvement. Dana records all decisions, assigns responsibilities for follow-up actions, and sets target deadlines. Review minutes are retained as a controlled record and referenced in subsequent improvement cycles.

9. Continuous Improvement Procedure

Continuous improvement is integrated into daily lab operations and formalized through regular root cause analysis of issues, corrective action logs, internal audits, and management reviews. Any staff member may propose improvements or report inconsistencies directly to Marcus Bell.

Recurring issues identified in certificate correction logs or customer complaints are flagged for detailed investigation. Marcus performs root cause analysis, consults with involved technicians, and documents results and proposed preventive actions. All corrective actions are monitored for timely closure, with process changes communicated via staff briefings or targeted training.

Suggestions for improvements that reduce errors, enhance documentation clarity, or increase throughput are discussed at team meetings, and the most promising are piloted and, if effective, rolled into standard practice. Larger changes or investments require Dana's review and approval.

The effectiveness of implemented improvements is measured through ongoing tracking of the same quality metrics established as objectives: correction rate, on-time delivery, and reference standard status. Data from these metrics is regularly reviewed to confirm continued progress or to identify the need for further intervention.